

LangChain & Agentic AI – From Basics to Advanced demo- 4 Notes

LangChain is a framework for building applications using Large Language Models.

company chatbot:

- take user's question

- create a proper prompt

- communicate with LLM

- search company documents

- remember conversation history

- call some external tools

- retrieve information

- process the result

- finally give answer

Many components are involved in this application.

Lang ---> refers Language Models

Chain --->Connect multiple components together

Question --->Retriever -->Documents -->Prompt --->Model --->
Answer

LangChain as an Orchestration Layer:

LangChain help us connect and coordinate the components of LLM application. Hence LC consider as Orchestration Layer for LLM Applications. The components can be:

Model

prompt

Memory/Histroy

Documents

Vector Database

Tools

Agents

etc

LLM = Engine

LangChain = System connecting and coordinating the parts around that engine.

LangChain as abstraction layer:

LangChain can provide an abstraction layer between our application and different underlying services

Abstraction vs Orchestration:

Abstraction: Hide or simplify complexity

Give an easy/common way to interact with different components

Orchestration : Coordinate multiple components

First prepare the prompt, then call the model, then parse the output

Abstraction simplifies. Orchestration coordinates.

LangChain can help us with both.

LangChain EcoSystem:

1. LangChain-->provides building blocks for developing LLM based applications
2. LangGraph-->Controlled and stateful workflows/agents
3. LangSmith-->Trace,debug and observability, monitoring and Evaluate LLM applications